

Project 2: Exploratory Data Analysis Report

Objective:

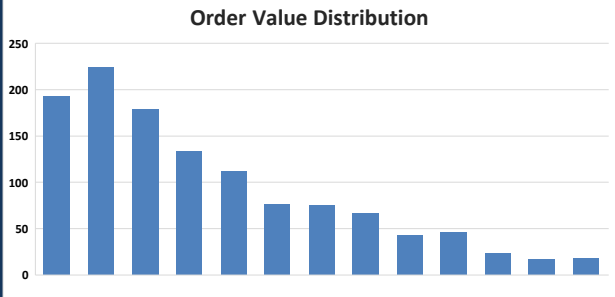
Analyze 1,200 customer orders to identify spending patterns, revenue drivers, and opportunities for business growth.

Data Overview:

- Total Orders: 1,200
- Dataset Type: Sales Transactions
- Analysis Focus: Spending Behavior & Revenue Patterns

Statistical Summary:

Metric	Value
Count	1200
Mean	1053.97
Median	823.62
Mode	635.90
Minimum	11.39
Maximum	3456.40
Standard Deviation	819.86



Key Insights:

1. Right-Skewed Spending Pattern

Mean exceeds Median, indicating that a small number of high-value purchases are influencing average revenue.

2. High Variability

A standard deviation of 819.86 suggests substantial variation in customer spending behavior.

3. Wide Revenue Range

Order values range from 11.39 to 3,456.40, highlighting both low-value and premium purchases.

4. Customer Spending Behavior

The distribution of order values is positively skewed, indicating that a small number of high-value orders significantly increase the overall average order value.

5. Product Performance

A small number of products contribute disproportionately to total revenue, highlighting key revenue-driving products.

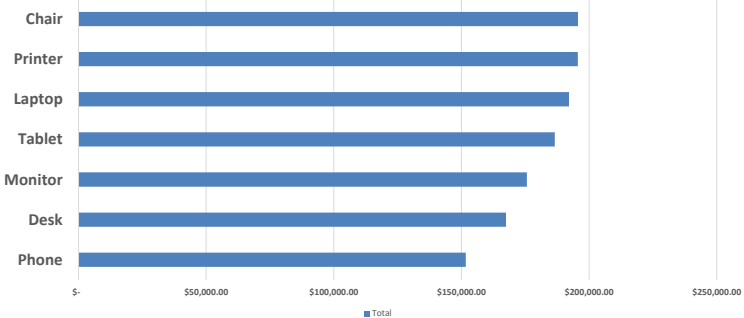
Recommendations:

- Identify characteristics of high-value customers and products.
- Maintain inventory levels for top-performing products.
- Develop targeted marketing campaigns for premium customer segments.

Conclusion:

The exploratory data analysis revealed valuable insights into customer spending behavior, product performance, and referral source effectiveness. The distribution of order values indicates that a relatively small number of high-value purchases contribute significantly to overall revenue. Additionally, the analysis identified key revenue-driving products and highlighted the impact of specific referral channels on business performance. These findings support data-driven decision-making for customer engagement, inventory management, and business growth.

Top Selling Products



Revenue by Referral Source

